Machine Learning Assignment – Deliverable 5.4

Recurrent Neural Network Based Classification

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1. Introduction

In this deliverable, recurrent neural network (RNN) based models were applied to the Human Activity Recognition (HAR) problem using the UCI HAR dataset. Unlike previous deliverables where handcrafted features were used, this approach works directly on raw time-series sensor data collected from smartphone accelerometers and gyroscopes. The objective of this deliverable is to analyze how sequential deep learning models learn temporal patterns from sensor signals and compare their performance with earlier approaches.

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2. Models Implemented

Three recurrent models were implemented and evaluated:

- **Simple RNN** was used as a baseline model to understand basic recurrent behavior.
- **Long Short-Term Memory (LSTM)** was selected due to its ability to retain long-term dependencies using memory cells.
- **Gated Recurrent Unit (GRU)** was implemented as a lightweight alternative to LSTM with fewer parameters and faster convergence.

All models were intentionally kept simple and trained for **20 epochs** to allow sufficient learning while maintaining clarity and interpretability.

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3. Training Observations

During training, both training and validation loss showed a generally decreasing trend for all models, indicating effective learning. Increasing the number of epochs from 10 to 20 resulted in noticeable performance improvements, particularly for LSTM and GRU models.

The Simple RNN showed unstable learning and limited improvement, whereas LSTM and GRU displayed smoother validation curves, demonstrating their ability to better capture temporal dependencies in sequential sensor data.

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4. Results and Discussion

Among the evaluated recurrent models, **GRU achieved the highest validation accuracy (~63%)**, followed by **LSTM (~60.79%)**, while the **Simple RNN showed poor performance (~32%)**.

Dynamic activities such as walking and walking upstairs were classified more accurately due to distinct motion patterns. Some confusion was observed among static activities like sitting, standing, and laying, which is expected because these activities generate very similar sensor signals.

Overall, the results highlight the effectiveness of gated recurrent architectures, with GRU slightly outperforming LSTM in this implementation.

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5. Comparison with Fully Connected Neural Network (Deliverable 5.3)

When compared with the Fully Connected Neural Network (FCNN) implemented in the previous deliverable, recurrent models achieved lower overall accuracy. The FCNN relied on handcrafted features, which summarized important characteristics of the signals, while recurrent models learned directly from raw sensor sequences.

Although recurrent models did not outperform feature-based approaches, they provided valuable insight into learning temporal dependencies directly from sensor data, which is essential for more advanced deep learning architectures.

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6. Limitations

Despite their advantages, recurrent models require longer training times and higher computational resources. Their performance is sensitive to hyperparameters such as the number of epochs, sequence length, and number of hidden units. More complex architectures such as stacked LSTMs, GRUs, or CNN–RNN hybrids could further improve performance but were avoided to keep the implementation simple and understandable.

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7. Comparison of All Approaches Used in the Project (Assignments 5.2–5.4)

Approach	Model	Input Representation	Best Accuracy	Strengths	Limitations
Feature Engineering + ML (5.2)	Random Forest	Handcrafted features	~93%	High accuracy, interpretable, fast training	Depends on manual feature extraction
Feature Engineering + DL (5.3)	FCNN	Handcrafted features	~90%	Simple neural architecture, stable learning	Ignores temporal sequence information
Raw Sequence Learning (5.4)	**GRU**	Raw sensor signals	**~63%**	Efficient recurrent model, stable learning	Requires more data and training
Raw Sequence Learning (5.4)	LSTM	Raw sensor signals	~60.79%	Captures temporal dependencies	Slightly lower accuracy than GRU
Raw Sequence Learning (5.4)	Simple RNN	Raw sensor signals	~32%	Conceptually simple	Vanishing gradient problem

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8. Conclusion

This deliverable successfully demonstrates the application of recurrent neural networks for time-series classification using raw sensor data. Among the evaluated models, **GRU achieved the best overall performance**, followed closely by LSTM, while Simple RNN showed limited capability in handling long temporal sequences. Although feature-based approaches achieved higher accuracy, recurrent models provide a strong foundation for learning temporal patterns directly from data. This work establishes a solid baseline for future exploration of more advanced deep learning architectures for human activity recognition